LLM-Driven Synthetic Text Generation for Privacy-Preserving Federated Learning

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***Abstract*—Federated learning (FL) enables collaborative fine-tuning of pre-trained language models (PLMs) across user devices by aggregating locally trained model parameters rather than raw data, exhibiting great potential in privacy-sensitive natural language processing (NLP) applications. However, the inherent tendency of model parameters to memorize training data puts existing efforts at risk of disclosing raw data. Knowledge distillation (KD) offers a privacy-preserving alternative for knowledge transfer between the cloud server and user devices by exchanging lightweight model predictions associated with a public proxy dataset. Nevertheless, a desirable proxy dataset may not always be available in practice as its construction requires careful deliberation and even prior knowledge about the private domain to ensure satisfactory performance, which conflicts with the privacy-preserving principles of FL. To tackle this challenge, this paper proposes FedSTG, a privacy-preserving FL framework empowered by synthetic text generation of large language models (LLMs), which combines the exceptional generation capability of a pre-trained LLM deployed on the cloud server with the domain-specific understanding capabilities of local PLMs distributed across user devices to generate high-quality synthetic text for knowledge transfer, facilitating the integration of KD and FL without the need for publicly available proxy data. Extensive experiments across a wide range of benchmark tasks demonstrate that FedSTG achieves superior performance without requiring any prior knowledge about the private domain, delivering a privacy-preserving FL solution for NLP.**

***Index Terms*—Federated learning, knowledge distillation, large language model, synthetic text generation.**

I. Introduction

RECENT advances in natural language processing (NLP) have been propelled by large-scale pre-trained language models (PLMs) [1], [2], [3], [4], [5], [6], [7], [8], [9] fine-tuned with vast amounts of domain-specific data. However, collecting user-generated textual data, which is inherently privacy-sensitive, from user devices to a cloud server for domain-specific fine-tuning often raises significant privacy concerns [10]. Federated learning (FL) [11] has emerged as a promising solution by delegating the fine-tuning process to individual devices and then gathering the model parameters locally trained on their respective private data at the cloud server for model aggregation. In this way, data privacy is ensured by transmitting only model parameters between the cloud server and user devices without exposing the underlying data [12], [13]. However, recent studies [14], [15] have revealed that the inherent tendency of model parameters to memorize training data [16], [17], which can be exploited by malicious attackers to infer sensitive information about user devices or even recover their raw data. As a result, FL remains vulnerable to privacy attacks despite the data never leaves the user device.

Knowledge distillation (KD) [18] offers a privacy-preserving alternative for knowledge transfer between the cloud server and user devices by exchanging lightweight model predictions rather than redundant model parameters. Specifically, the cloud server and user devices share a public proxy dataset as the carrier for knowledge transfer, on which each device calculates the model predictions using its locally trained PLM and uploads them to the cloud server. The cloud server then aggregates the model predictions received from individual devices into ensemble predictions, which are used to train a global PLM by minimizing the discrepancy between its model predictions and the ensemble predictions on the public proxy dataset. Meanwhile, each device can learn knowledge from peer devices by incorporating the ensemble predictions with the public proxy dataset to further refine its local PLM. In this way, the risk of privacy leakage in FL is significantly reduced due to sharing model predictions exposes much less sensitive information than sharing model parameters.

However, incorporating KD with FL is non-trivial due to its reliance on a public proxy dataset, whose relevance to the private domain is crucial for effective knowledge transfer. To this end, most existing approaches [19], [20] assume that public proxy data and private local data originate from the same underlying domain, which inevitably exposes private local data to potential privacy threats. More recent efforts [21], [22] relax this assumption by sampling cross-domain proxy data from another relevant domain. Nevertheless, obtaining a desirable proxy dataset in practice remains challenging as its construction requires careful deliberation and even prior knowledge about the private domain to ensure satisfactory performance, which conflicts with the privacy-preserving principles of FL. A real-world use case occurs in cross-institutional collaboration, where clinical institutions are legally prohibited from sharing real patient records. Relying on public clinical corpora for cross-institutional knowledge transfer leads to severe domain mismatch as it fails to faithfully capture the nuanced

Received 9 July 2025; revised 9 December 2025; accepted 16 January 2026. Date of publication 26 January 2026; date of current version 4 February 2026. This work was supported by the National Natural Science Foundation of China (NSFC) under Grant 61966038 and Grant 62266051. The associate editor coordinating the review of this article and approving it for publication was Prof. Mark Hasegawa-Johnson.*(Corresponding author: Jin Wang.)*

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The code for this paper is publicly available at: https://github.com/maxinge8698/FedSTG

Digital Object Identifier 10.1109/TASLPRO.2026.3657644

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distributional patterns and domain-specific terminologies contained in private clinical corpora across institutions. In contrast, curating a new clinical dataset requires institutions to disclose their domain characteristics that potentially leak sensitive information. In this case, synthetic clinical data serves as a promising alternative to public clinical data for knowledge transfer while stringently adhering to data protection requirements.

To tackle this challenge, this paper proposes FedSTG, a privacy-preserving FL framework empowered by synthetic text generation of large language models (LLMs), which combines the exceptional generation capability of a pre-trained LLM deployed on the cloud server with the domain-specific understanding capabilities of local PLMs distributed across user devices to generate high-quality synthetic text for knowledge transfer, facilitating the integration of KD and FL without the need for publicly available proxy data.

In summary, the main contributions made in this paper are organized as follows:

- We propose FedSTG, a privacy-preserving FL framework that facilitates knowledge transfer between the cloud server and user devices by sharing lightweight model predictions associated with synthetic data, thereby minimizing the risk of privacy leakage in FL.
- We develop an LLM-driven synthetic text generation strategy that equips the cloud server with a pre-trained LLM as the synthetic text generator and then incorporates local PLMs distributed across user devices as evaluators to facilitate the generation of domain-specific synthetic text, eliminating the need to curate a public proxy dataset.
- We conduct extensive experiments on a wide range of benchmark tasks, and the empirical results demonstrate that FedSTG achieves superior performance compared to existing KD-based FL frameworks without requiring any prior knowledge about the private domain.

The remainder of this paper is organized as follows. Section II reviews the relevant literature. Section III describes the proposed FedSTG framework. Section IV details the experimental setup and presents empirical results. Section V concludes this study with a brief summary and discusses potential limitations for future research.

II. Related Work

A. Privacy-Preserving Federated Learning

FL enables multiple user devices holding private data to collaborate with the cloud server in fine-tuning a global PLM without compromising their data privacy, exhibiting great potential in privacy-sensitive NLP applications [23], [24], [25], [26]. However, the iterative communication of model parameters between the cloud server and user devices makes FL vulnerable to privacy attacks, which is further exacerbated by the massive parameter scale of PLMs [27], [28]. To prevent the leakage of sensitive information, a series of efforts on privacy-preserving FL have recently been undertaken. Basu et al. [29] investigated the impact of differential privacy (DP) [30], [31] on the federated fine-tuning of PLMs under varying levels of privacy guarantees, and empirically suggested that incorporating DP with FL leads to significant utility degeneration due to the considerable noise injected into the model parameters. Deng et al. [32] proposed SEFL, a secure and efficient FL framework that employs two non-colluding servers to defend against potential attacks from an untrusted server during the aggregation process, where the aggregation server is responsible for the aggregation of encrypted local model updates while the cryptography server is in charge of the decryption key. However, implementing encryption and decryption for model parameters incurs substantial computational overhead and communication latency. Zhang et al. [33] proposed FedeVocab, a federated model decomposition framework that protects the mappings between sensitive words and their associated embeddings on user devices from data reconstruction attacks by detaching the local embedding parameters from federated aggregation. However, the mismatch between the local embedding layer and the global encoder layer results in non-negligible performance degradation. Nagy et al. [34] proposed a privacy-preserving FL framework that utilizes a rolling-hash-based embedding approach to privatize textual features and a randomized-response-based perturbation technique to randomize locally quantized model updates. However, the combination of multiple privacy-preserving techniques comes with significant performance sacrifices.

B. Knowledge Distillation for Federated Learning

To overcome these bottlenecks, Sui et al. [19] proposed FedED, a KD-based FL framework that transfers knowledge from the ensemble of local PLMs to the global PLM by encouraging them to make similar predictions on a public proxy dataset. By exchanging lightweight model predictions rather than redundant model parameters, the volume of messages transmitted in each communication round is dramatically reduced, leading to less exposure of sensitive information. However, partitioning the public proxy dataset from labeled data within the private domain inevitably divulges private local data to some extent. Lin et al. [20] proposed FedDF, which leverages unlabeled data from the private domain to relax constraints on data availability, bringing only limited improvements in mitigating privacy concerns. Ma et al. [21] proposed FedID, which explores the feasibility of leveraging cross-domain unlabeled data for knowledge transfer. However, it requires the cloud server to retain a small amount of labeled data for correction of confirmation bias to ensure satisfactory performance. Wang et al. [22] introduced a pre-trained LLM to promote privacy-preserving FL by carefully handpicking data similar to the private domain from publicly available data sources through distribution matching. However, a desirable proxy dataset may not always be available in practice. Wu et al. [35] suggested that well-designed prompts can guide a pre-trained LLM to generate synthetic data close to the private domain without accessing the underlying private data. However, constructing such prompts requires careful deliberation and even prior knowledge about the private domain, which conflicts with the privacy-preserving principles of FL.

More recent studies [36], [37], [38], [39], [40] have demonstrated that data-free knowledge distillation (DFKD) [41], [42] offers a novel paradigm for privacy-preserving FL by

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equipping the cloud server with a generative adversarial network (GAN) [43], [44], [45] to directly generate domain-specific synthetic images without requiring any prior knowledge about the private domain. Unfortunately, these frameworks are designed for computer vision tasks and thus fail to be readily extended to NLP due to discrete input spaces and large foundation models. To bridge this gap, this work aims to equip the cloud server with a pre-trained LLM as the synthetic text generator and then design a collaborative prompting strategy to guide it in generating domain-specific synthetic text for knowledge transfer without requiring any prior knowledge about the private domain, delivering a privacy-preserving FL solution for NLP.

III. Methods

A. Problem Formulation

Given a cloud server and K user devices collaborating on a C-class text classification task $\mathcal{T} = \{\mathcal{X}, \mathcal{Y}\}$, where $\mathcal{X}$ and $\mathcal{Y}$ represent input and output spaces, respectively. Each device $k \in \{1, 2, \ldots, K\}$ locally stores a domain-specific private dataset $\mathcal{D}_k = \{(x_k^{(i)}, y_k^{(i)}) | i = 1, 2, \ldots, N_k\}$ and hosts a local PLM $\mathcal{F}_k$ parameterized by Θ_k, where $x_k^{(i)} \in \mathcal{X}$ represents the input text sequence, $y_k^{(i)} \in \mathcal{Y}$ represents the corresponding class label, and N_k is the amount of private local data. The goal is to collaboratively fine-tune a global PLM $\mathcal{F}$ parameterized by Θ on the cloud server using domain-specific knowledge from user devices without compromising their data privacy.

FL presents a straightforward approach by distributing the parameters of the global PLM to individual devices as the initialization of their local PLMs, and then aggregating the model parameters locally trained on their respective private dataset at the cloud server. However, the exposure of model parameters increases the risk of disclosing private local data due to their tendency to memorize training data.

KD-based FL significantly mitigates privacy risks by replacing the aggregation of model parameters with KD, which only requires the cloud server and user devices to exchange their model predictions associated with a public proxy dataset for knowledge transfer. Nevertheless, a desirable proxy dataset $\mathcal{D}_{pub} = \{x_{pub}^{(i)} | i = 1, 2, \ldots, N_{pub}\}$ may not always be available in practice, which challenges the application of KD for privacy-preserving FL. The proposed FedSTG framework aims to tackle this challenge by sharing model predictions associated with a synthetic dataset $\mathcal{D}_{syn} = \{\hat{x}^{(i)} | i = 1, 2, \ldots, N_{syn}\}$. An overview of the proposed FedSTG framework is illustrated in Fig. 1.

B. LLM-Driven Synthetic Text Generation

Inspired by the exceptional generation capabilities of LLMs, which are capable of generating coherent synthetic text on par with human-generated text owing to the comprehensive linguistic knowledge accumulated by extensive pre-training on large-scale open-ended corpora from diverse sources, serving as highly promising synthetic text generators [46], [47]. This motivates us to explore an LLM-driven synthetic text generation strategy for the generation of domain-specific synthetic text, facilitating knowledge transfer between the cloud server and user devices without the need for publicly available proxy data.

To this end, the cloud server is equipped with a pre-trained LLM $\mathcal{G}$ parameterized by $\Theta_{\mathcal{G}}$ as the underlying synthetic text generator. At each communication round $t \in \{1, 2, \ldots, T\}$, the cloud server initializes the LLM $\mathcal{G}$ with the parameters $\Theta_{\mathcal{G}}^{t-1}$ from the previous round $t-1$, as follows:

$$\Theta_{\mathcal{G}}^{t} \leftarrow \Theta_{\mathcal{G}}^{t-1}, \tag{1}$$

and samples a class label $\hat{y}^{(i)}$ from a uniform distribution $\mathcal{U}(0, C-1)$, *e.g.*, "0", which is further mapped by a verbalizer $\mathcal{V}(\cdot)$ to the corresponding label word $\mathcal{V}(\hat{y}^{(i)})$ in the vocabulary space V, *e.g.*, "*negative*". Subsequently, a predefined template $\mathcal{M}(\cdot)$ is used to wrap it into a label-descriptive prompt $\mathcal{M}(\mathcal{V}(\hat{y}^{(i)}))$, *e.g.*, "*Please generate a negative movie review.*", which can be naturally used to prompt the LLM $\mathcal{G}$ to generate a synthetic text $\hat{x}^{(i)}$ conditioned on the desired attribute $\hat{y}^{(i)}$, as follows:

$$\hat{x}^{(i)} = \mathcal{G}\left(\mathcal{M}\left(\mathcal{V}\left(\hat{y}^{(i)}\right)\right); \Theta_{\mathcal{G}}^{t}\right). \tag{2}$$

The generation of $\hat{x}^{(i)}$ involves autoregressive sampling of the next token $\hat{x}_l^{(i)} \in V$ from the output probability distribution $p_{\Theta_{\mathcal{G}}^t}(\hat{x}_l^{(i)} | \mathcal{M}(\mathcal{V}(\hat{y}^{(i)})), \hat{x}_{<l}^{(i)})$ based on the control variable $\mathcal{M}(\mathcal{V}(\hat{y}^{(i)}))$ and preceding tokens $\hat{x}_{<l}^{(i)} = \{\hat{x}_1^{(i)}, \hat{x}_2^{(i)}, \ldots, \hat{x}_{l-1}^{(i)}\}$ as the context, as follows:

$$\begin{aligned} & p_{\Theta_{\mathcal{G}}^t}\left(\hat{x}^{(i)} | \mathcal{M}\left(\mathcal{V}\left(\hat{y}^{(i)}\right)\right)\right) \\ & = \prod_{l=1}^{n^{(i)}} p_{\Theta_{\mathcal{G}}^t}\left(\hat{x}_l^{(i)} | \mathcal{M}\left(\mathcal{V}\left(\hat{y}^{(i)}\right)\right), \hat{x}_{<l}^{(i)}\right), \end{aligned} \tag{3}$$

where $p_{\Theta_{\mathcal{G}}^t}(\hat{x}^{(i)} | \mathcal{M}(\mathcal{V}(\hat{y}^{(i)})))$ represents the likelihood that the LLM $\mathcal{G}$ generates the text sequence $\hat{x}^{(i)}$ containing $n^{(i)}$ tokens. Furthermore, various decoding strategies such as top-k sampling [48] and nucleus sampling (*a.k.a.* top-p sampling) [49] can be adopted during generation to increase the diversity of generated synthetic text. The above generation process is repeated for N_{syn} times to construct a synthetic dataset $\mathcal{D}_{syn} = \{\hat{x}^{(i)} | i = 1, 2, \ldots, N_{syn}\}$ for knowledge transfer between the cloud server and user devices.

However, merely prompting the LLM $\mathcal{G}$ with task and label descriptions tends to generate highly repetitive or even irrelevant synthetic text, which is far from the private domain due to the lack of domain-specific knowledge [50], [51], [52], [53], [54], [55]. To navigate the LLM $\mathcal{G}$ to generate domain-specific synthetic text for task $\mathcal{T}$, motivated by recent advances in fine-tuning with feedback to align language models [56], we propose a collaborative prompting strategy that incorporates the locally trained PLMs $\{\mathcal{F}_k | k \in \mathcal{K}^t\}$ distributed across user devices as evaluators to provide rewards on the quality of the generated synthetic text, aiming to enrich the LLM $\mathcal{G}$ with their domain-specific knowledge and thus enhance its generation capability within the same domain.

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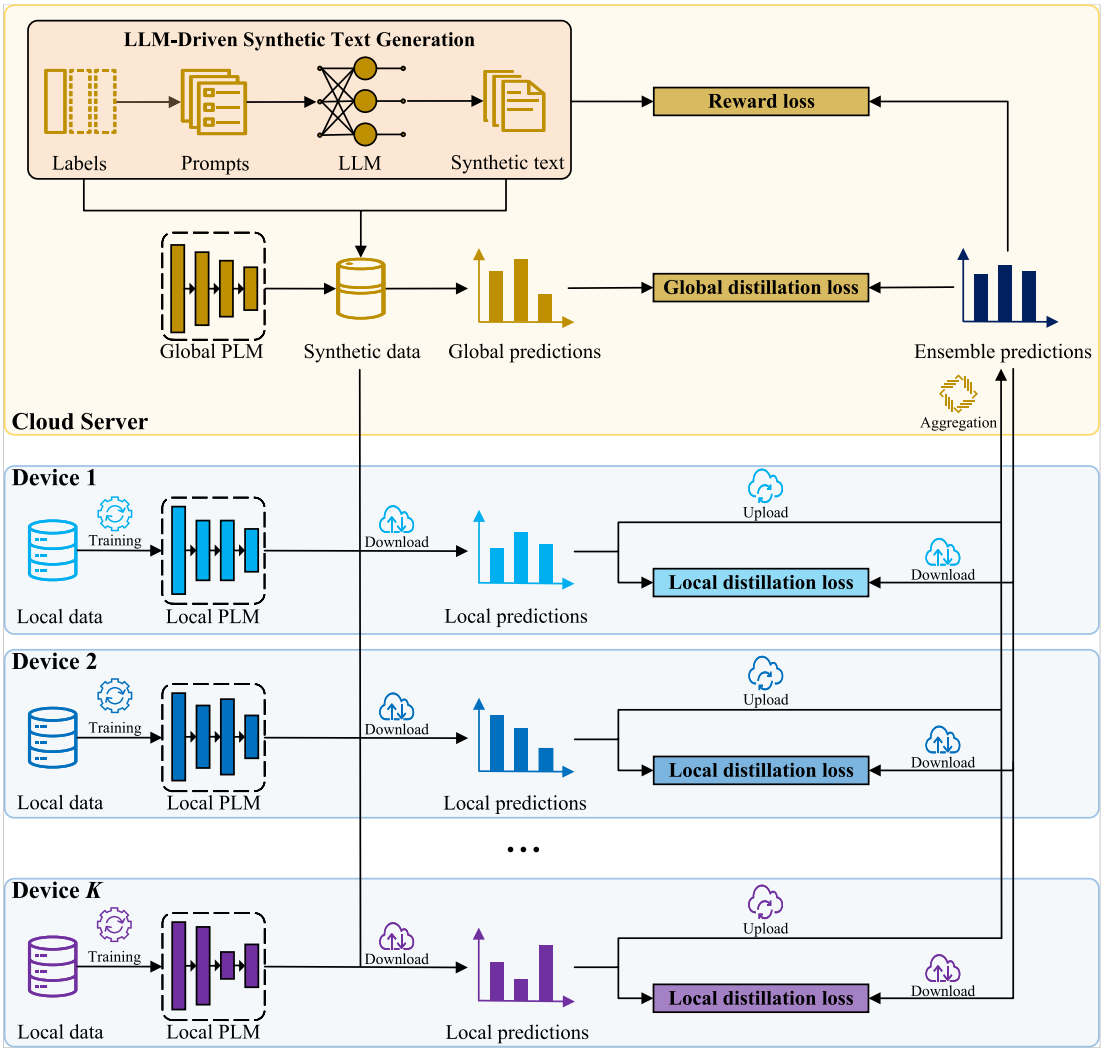

Fig. 1. Illustration of the proposed FedSTG framework.

Specifically, the synthetic dataset $\mathcal{D}_{syn}$ generated in each round are distributed to a set of available devices $\mathcal{K}^t \subseteq \{1, 2, \ldots, K\}$ sampled from all devices based on the sampling fraction q, where $|\mathcal{K}^t| = q \cdot K$ aiming to simulate real-world scenarios where not all devices are always online or available. Subsequently, each device $k \in \mathcal{K}^t$ initializes its local PLM $\mathcal{F}_k$ with the parameters Θ_k^{t-1} from the previous round $t-1$, as follows:

$$\Theta_k^t \leftarrow \Theta_k^{t-1}, \tag{4}$$

and then locally trains it to extract domain-specific knowledge from its local dataset $\mathcal{D}_k$, as follows:

$$\mathcal{L}_k^{\mathrm{CE}}(\mathcal{D}_k; \Theta_k^t) = \frac{1}{N_k} \sum_{i=1}^{N_k} \ell_{\mathrm{CE}} \left(y_k^{(i)}, \mathcal{F}_k \left(x_k^{(i)}; \Theta_k^t \right) \right), \tag{5}$$

where $\ell_{\mathrm{CE}}(\cdot)$ is the cross-entropy loss to measure the classification error. Once completed, the model predictions of the locally trained PLM $\mathcal{F}_k$ on the synthetic dataset $\mathcal{D}_{syn}$ are calculated and then uploaded to the cloud server, as follows:

$$\hat{Y}_k^t = \{\mathcal{F}_k(\hat{x}^{(i)}; \Theta_k^t) | i = 1, 2, \ldots, N_{syn}\}. \tag{6}$$

To mitigate the impact of any single under-trained or biased device, the cloud server aggregates the received local model predictions into ensemble predictions based on the proportion of private local data held by each device to deliver more reliable predictions, as follows:

$$\hat{Y}^t = \left\{ \sum_{k \in \mathcal{K}^t} \frac{N_k}{N} \mathcal{F}_k(\hat{x}^{(i)}; \Theta_k^t) | i = 1, 2, \ldots, N_{syn} \right\}, \tag{7}$$

where $N = \sum_{k \in \mathcal{K}^t} N_k$ is the cumulative amount of private local data across all participating devices $\mathcal{K}^t$. Subsequently, the cloud server calculates a reward score for each synthetic sample based on the corresponding ensemble prediction, as follows:

$$r\left(\hat{x}^{(i)}, \hat{y}^{(i)}\right) = \left(\sum_{k \in \mathcal{K}^t} \frac{N_k}{N} \mathcal{F}_k(\hat{x}^{(i)}; \Theta_k^t) \right)_{\hat{y}^{(i)}}$$

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$$-\left(1-\left(\sum_{k\in\mathcal{K}^t}\frac{N_k}{N}\mathcal{F}_k\left(\hat{x}^{(i)};\Theta_k^t\right)\right)_{\hat{y}^{(i)}}\right), \tag{8}$$

where the first term $(\sum_{k\in\mathcal{K}^t}\frac{N_k}{N}\mathcal{F}_k(\hat{x}^{(i)};\Theta_k^t))_{\hat{y}^{(i)}}$ represents the probability that the ensemble of local PLMs classifies sample $\hat{x}^{(i)}$ into the target class label $\hat{y}^{(i)}$, while the last term $1-(\sum_{k\in\mathcal{K}^t}\frac{N_k}{N}\mathcal{F}_k(\hat{x}^{(i)};\Theta_k^t))_{\hat{y}^{(i)}}$ represents the cumulative probability classified into all other class labels. In this way, the reward score $r(\hat{x}^{(i)},\hat{y}^{(i)})$ transforms the probability space from [0,1] to a symmetric range of $[-1,1]$ for numerical evaluation on the quality of synthetic sample $\hat{x}^{(i)}$. Accordingly, the LLM $\mathcal{G}$ is fine-tuned using the reward loss to improve its domain-specific generation capability, as follows:

$$\begin{aligned}\mathcal{L}_{\mathcal{G}}^{\mathrm{RW}}\left(\mathcal{D}_{syn};\Theta_{\mathcal{G}}^t\right) &= \frac{1}{N_{syn}}\sum_{i=1}^{N_{syn}} -r\left(\hat{x}^{(i)},\hat{y}^{(i)}\right)\cdot \\ &\quad\left(\log p_{\Theta_{\mathcal{G}}^t}\left(\hat{x}^{(i)}|\mathcal{M}(\mathcal{V}(\hat{y}^{(i)}))\right)\right. \\ &\quad\left.-\log p_{\Theta_{\mathcal{G}}^{t-1}}\left(\hat{x}^{(i)}|\mathcal{M}\left(\mathcal{V}(\hat{y}^{(i)})\right)\right)\right),\end{aligned} \tag{9}$$

where $p_{\Theta_{\mathcal{G}}^t}(\hat{x}^{(i)}|\mathcal{M}(\mathcal{V}(\hat{y}^{(i)})))$ and $p_{\Theta_{\mathcal{G}}^{t-1}}(\hat{x}^{(i)}|\mathcal{M}(\mathcal{V}(\hat{y}^{(i)})))$ represent the likelihood that the LLM $\mathcal{G}$ generates the text sequence $\hat{x}^{(i)}$ given the current parameters $\Theta_{\mathcal{G}}^t$ and the frozen initial parameters $\Theta_{\mathcal{G}}^{t-1}$, respectively. Notably, when the ensemble probability of classifying $\hat{x}^{(i)}$ to $\hat{y}^{(i)}$ exceeds 0.5, the reward score $r(\hat{x}^{(i)},\hat{y}^{(i)})$ will be positive, and minimizing the reward loss $\mathcal{L}_{\mathcal{G}}^{\mathrm{RW}}$ will make $\log p_{\Theta_{\mathcal{G}}^t}(\hat{x}^{(i)}|\mathcal{M}(\mathcal{V}(\hat{y}^{(i)})))$ larger than $\log p_{\Theta_{\mathcal{G}}^{t-1}}(\hat{x}^{(i)}|\mathcal{M}(\mathcal{V}(\hat{y}^{(i)})))$, which means the probability that the LLM $\mathcal{G}$ generates $\hat{x}^{(i)}$ is increased. Otherwise, the reward score $r(\hat{x}^{(i)},\hat{y}^{(i)})$ will be negative, and minimizing the reward loss $\mathcal{L}_{\mathcal{G}}^{\mathrm{RW}}$ will make $\log p_{\Theta_{\mathcal{G}}^t}(\hat{x}^{(i)}|\mathcal{M}(\mathcal{V}(\hat{y}^{(i)})))$ smaller than $\log p_{\Theta_{\mathcal{G}}^{t-1}}(\hat{x}^{(i)}|\mathcal{M}(\mathcal{V}(\hat{y}^{(i)})))$, which means the probability that the LLM $\mathcal{G}$ generates $\hat{x}^{(i)}$ is decreased. As a result, the design of the reward loss $\mathcal{L}_{\mathcal{G}}^{\mathrm{RW}}$ encourages the LLM $\mathcal{G}$ to generate high-quality synthetic text while discouraging the generation of low-quality synthetic text.

C. Domain-Specific Knowledge Transfer

Subsequently, the cloud server can leverage these synthetic samples and the corresponding ensemble predictions to fine-tune the global PLM $\mathcal{F}$ through KD, learning domain-specific knowledge from user devices, as follows:

$$\begin{aligned}&\mathcal{L}^{\mathrm{KD}}\left(\mathcal{D}_{syn};\Theta^t\right)\\ &=\frac{1}{N_{syn}}\sum_{i=1}^{N_{syn}}\ell_{KL}\left(\sum_{k\in\mathcal{K}^t}\frac{N_k}{N}\mathcal{F}_k(\hat{x}^{(i)};\Theta_k^t)||\mathcal{F}(\hat{x}^{(i)};\Theta^t)\right),\end{aligned} \tag{10}$$

where $\ell_{\mathrm{KL}}(\cdot||\cdot)$ is the KL-divergence loss to measure the discrepancy between two distributions. Note that the LLM $\mathcal{G}$ and the global PLM $\mathcal{F}$ are trained alternately on the cloud server without imposing additional storage requirements and computational burdens on resource-constrained user devices. Meanwhile, the ensemble predictions are redistributed to each device $k\in\mathcal{K}^t$, which incorporates the received ensemble predictions to further refine its local PLM $\mathcal{F}_k$ on these synthetic samples through KD, as follows:

$$\begin{aligned}&\mathcal{L}_k^{\mathrm{KD}}(\mathcal{D}_{syn};\Theta_k^t)\\ &=\frac{1}{N_{syn}}\sum_{i=1}^{N_{syn}}\ell_{KL}\left(\sum_{k\in\mathcal{K}^t}\frac{N_k}{N}\mathcal{F}_k(\hat{x}^{(i)}|\Theta_k^t)||\mathcal{F}_k(\hat{x}^{(i)};\Theta_k^t)\right).\end{aligned} \tag{11}$$

In the next round of communication, the newly generated synthetic dataset will replace the one generated in the previous round without incurring additional storage requirements on resource-constrained user devices. Through multi-round refinement during federated training, the quality of synthetic text progressively improves as local PLMs are iteratively aligned, serving as a high-quality data source for domain-specific knowledge transfer between the cloud server and user devices without requiring any prior knowledge about the private domain. The complete training procedures of the proposed FedSTG framework are outlined in Algorithm 1.

IV. Experiments

A. Datasets

The experimental evaluation is conducted on three types of natural language understanding tasks from the General Language Understanding Evaluation (GLUE) benchmark [57], as follows:

- *Sentiment Analysis:* The Stanford Sentiment Treebank (SST-2) [58] is a single-sentence binary classification task, aiming to determine whether a given sentence expresses a positive or negative sentiment. The dataset consists of 67,349 training samples and 872 development samples, with accuracy (Acc) as the evaluation metric.
- *Paraphrase Detection:* The Quora Question Pairs (QQP)[1] is a sentence-pair binary classification task, aiming to determine whether two given sentences are semantically equivalent or not. The dataset consists of 363,846 training samples and 40,430 development samples, with F1-score (F1) and accuracy (Acc) as evaluation metrics.
- *Natural Language Inference:* The Multi-genre Natural Language Inference (MNLI) [59] is a sentence-pair ternary classification task, aiming to determine whether a given hypothesis sentence is entailed, neutral, or contradictory with respect to a given premise sentence. The dataset consists of 392,702 training samples, and the development set is divided into a matched version involving 9,815 samples and a mismatched version involving 9,832 samples, with matched accuracy (Acc m) and mismatched accuracy (Acc mm) as evaluation metrics, respectively.

To simulate non-independent and identically distributed (non-IID) private local data across user devices, the training set of each dataset serves as the private dataset and is partitioned into

[1]https://quoradata.quora.com/First-Quora-Dataset-Release-Question-Pairs

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Algorithm 1: Privacy-Preserving Federated Learning Empowered by Synthetic Text Generation (FedSTG).

Input: K user devices with local datasets $\{\mathcal{D}_k\}_{k=1}^K$ and local PLMs $\{\mathcal{F}_k\}_{k=1}^K$ parameterized by $\{\Theta_k^0\}_{k=1}^K$, a cloud server with a global PLM $\mathcal{F}$ parameterized by Θ^0 and a pre-trained LLM $\mathcal{G}$ parameterized by $\Theta_{\mathcal{G}}^0$, number of communication rounds T, sampling fraction q, number of synthetic samples N_{syn}, learning rates η_k, η, and $\eta_{\mathcal{G}}$

Output: The global PLM $\mathcal{F}$ parameterized by Θ^T

for each communication round $t = 1, 2, \ldots, T$ **do**
  The cloud server initializes the global PLM parameters $\Theta^t \leftarrow \Theta^{t-1}$
  The cloud server initializes the pre-trained LLM parameters $\Theta_{\mathcal{G}}^t \leftarrow \Theta_{\mathcal{G}}^{t-1}$
  The cloud server generates a synthetic dataset $\mathcal{D}_{syn} = \{\hat{x}^{(i)}, |i = 1, 2, \ldots, N_{syn}\}$ by prompting the pre-trained LLM $\mathcal{G}$ with task and label descriptions
  $m \leftarrow q \cdot K$
  $\mathcal{K}^t \leftarrow$ (random set of m devices)
  The cloud server broadcasts the synthetic dataset $\mathcal{D}_{syn}$ to participating devices $\mathcal{K}^t$
  for each participating device $k \in \mathcal{K}^t$ *in parallel* **do**
    Initialize the local PLM parameters $\Theta_k^t \leftarrow \Theta_k^{t-1}$
    Update the local PLM via $\Theta_k^t \leftarrow \Theta_k^t - \eta_k \nabla \mathcal{L}_k^{\mathrm{CE}}(\mathcal{D}_k; \Theta_k^t)$
    Calculate local predictions $\hat{Y}_k^t = \{\mathcal{F}_k(\hat{x}^{(i)}; \Theta_k^t) | i = 1, 2, \ldots, N_{syn}\}$
    Upload local predictions $\hat{Y}_k^t$ to the cloud server
  end for
  The cloud server aggregates local predictions $\{\hat{Y}_k^t\}_{k \in \mathcal{K}^t}$ into ensemble predictions $\hat{Y}^t = \{\sum_{k \in \mathcal{K}^t} \frac{N_k}{N} \mathcal{F}_k(\hat{x}^{(i)}; \Theta_k^t) | i = 1, 2, \ldots, N_{syn}\}$
  The cloud server updates the global PLM via $\Theta^t \leftarrow \Theta^t - \eta \nabla \mathcal{L}^{\mathrm{KD}}(\mathcal{D}_{syn}; \Theta^t)$.
  The cloud server incorporates LoRA to update the pre-trained LLM via $\Theta_{\mathcal{G}}^t \leftarrow \Theta_{\mathcal{G}}^t - \eta_{\mathcal{G}} \nabla \mathcal{L}_{\mathcal{G}}^{\mathrm{RW}}(\mathcal{D}_{syn}; \Theta_{\mathcal{G}}^t)$
  The cloud server broadcasts ensemble predictions $\hat{Y}^t$ to participating devices $\mathcal{K}^t$
  for each participating device $k \in \mathcal{K}^t$ *in parallel* **do**
    Update the local PLM via $\Theta_k^t \leftarrow \Theta_k^t - \eta_k \nabla \mathcal{L}_k^{\mathrm{KD}}(\mathcal{D}_{syn}; \Theta_k^t)$
  end for
end for
return Θ^T

K isolated silos using the Dirichlet distribution $\mathrm{Dir}(\alpha)$ [60]. The resulting sub-dataset $\mathcal{D}_k \sim \mathrm{Dir}(\alpha)$ is allocated to the k-th device as its local dataset, where the concentration parameter α determines the degree of non-IID characteristics, and a smaller value results in a higher shift in quantity and label distributions across user devices.

Furthermore, we introduce a cross-domain dataset for each benchmark task to act as the public proxy dataset required by existing KD-based FL frameworks for compatible comparisons with them. For SST-2, we adopt the training set of the Internet Movie Database (IMDB) dataset [61] as the public proxy dataset, which consists of 25,000 samples. For QQP, we adopt the training set of the Paraphrase Adversaries from Word Scrambling (PAWS) dataset [62] as the public proxy dataset, which consists of 49,401 samples. For MNLI, we adopt the training set of the Stanford Natural Language Inference (SNLI) dataset [63] as the public proxy dataset, which consists of 550,152 samples.

B. Baselines

To evaluate the effectiveness of the proposed FedSTG framework, we conduct a comprehensive comparison with five types of existing KD-based FL frameworks, as follows:

- *FedED:* FedED [19] relies on a *labeled public dataset* to transfer knowledge from user devices to the cloud server without uploading local model parameters. However, the reverse knowledge transfer from the cloud server to user devices still involves downloading the parameters of the global PLM for the initialization of their local PLMs, thereby supporting only homogeneous FL scenarios.
- *MHAT:* MHAT [64] can be viewed as a privacy-preserving extension of FedED that enables mutual knowledge transfer between the cloud server and user devices without sharing any model parameters, thereby supporting both heterogeneous and heterogeneous FL scenarios.
- *FedDF:* FedDF [20] relies on an *unlabeled public dataset* to further fine-tune the preliminary global PLM aggregated from local model parameters using the ensemble knowledge of local PLMs, thereby supporting only homogeneous FL scenarios.
- *DS-FL:* DS-FL [65] can be viewed as a privacy-preserving extension of FedDF that enables directional knowledge transfer between the cloud server and user devices without sharing any model parameters, thereby supporting both homogeneous and heterogeneous FL scenarios.
- *FedID:* FedID [21] further relaxes the constraint of DS-FL on data availability to a *cross-domain unlabeled public dataset*. However, it requires the cloud server to retain a *small-scale labeled dataset*[2] to bridge the domain gap between private local data and public proxy data.

Instead, FedSTG eliminates the dependence of DS-FL on a public proxy dataset by equipping the cloud server with a pre-trained LLM to directly generate a *domain-specific synthetic dataset* for knowledge transfer between the cloud server and user devices.

C. Implementation Details

We conduct the experiments under the FL setup with a cloud server and $K = 10$ user devices, where the sampling fraction is set to $q = 1$, and the number of communication rounds is set to $T = 10$. The concentration parameter is set to $\alpha = 100$ and

[2]For a fair comparison with FedID, we adopt the development sets of IMDB, PAWS, and SNLI as the small-scale labeled datasets held by the cloud server for SST-2, QQP, and MNLI, respectively, which consist of 25,000, 8,000, and 10,000 samples, respectively.

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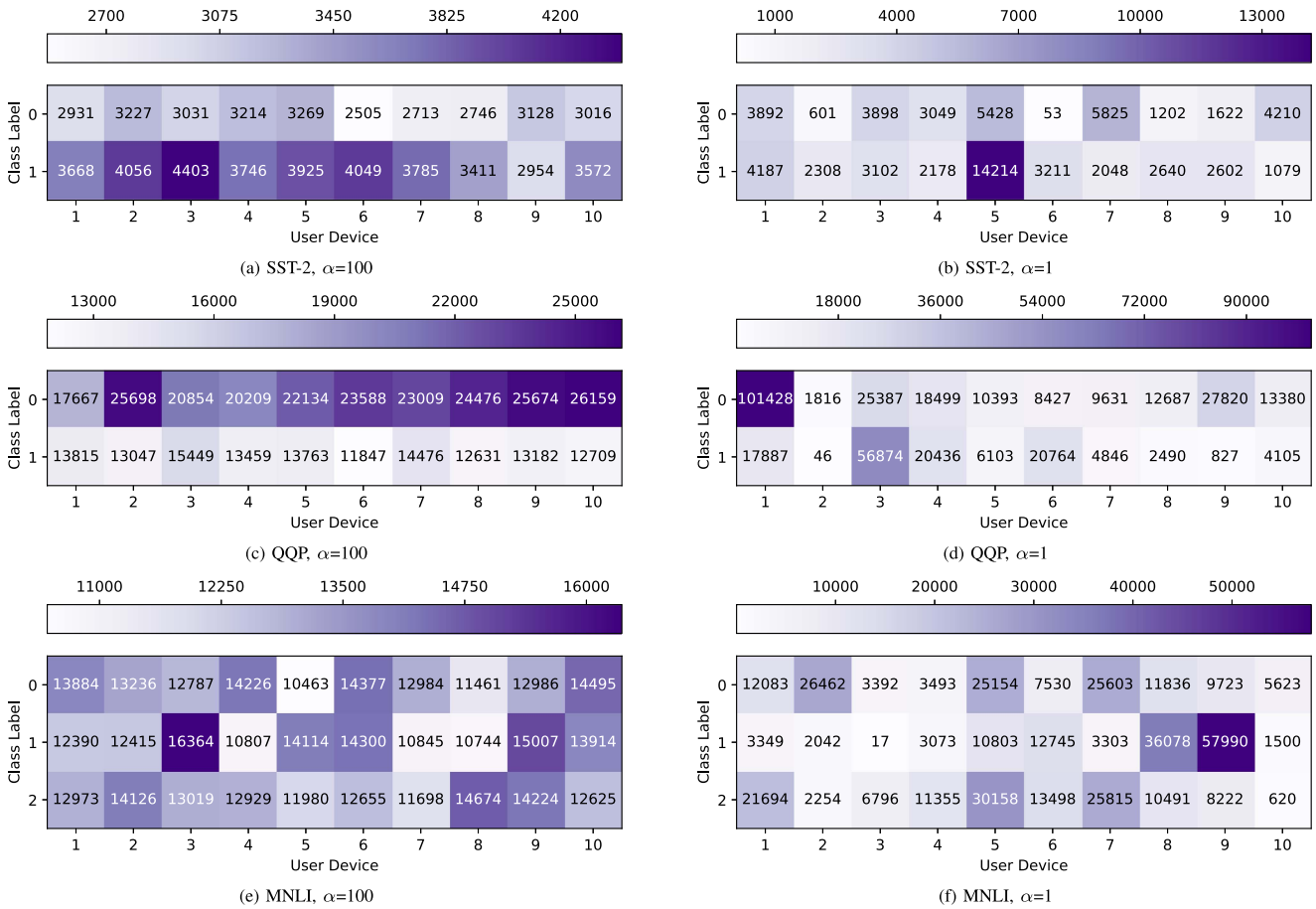

Fig. 2. Visualization of data heterogeneity across user devices on benchmark datasets.

$\alpha = 1$ for the simulation of IID and non-IID data distributions, respectively, and the resulting data distributions across user devices on benchmark tasks are shown in Fig. 2. In homogeneous FL scenarios where user devices are constrained to replicate the identical model architecture as the cloud server, the global PLM and all local PLMs are configured as BERT-base [1] for evaluation. In heterogeneous FL scenarios where user devices are allowed to customize model architectures according to their data characteristics and resource constraints, the global PLM is configured as BERT-base while local PLMs are assigned to either BERT-base or DistilBERT [66] at a hybrid ratio of 1:1 to simulate varying scales of PLMs for evaluation. Unless otherwise specified, the experiments are conducted under homogeneous FL scenarios with $\alpha = 100$, and the pre-trained LLM is configured as Llama-3.1-8B-Instruct [9] by default. The prompt templates and generation configurations on different benchmark tasks are listed in Table I.

For each benchmark task, the batch size is set to 16, the maximum sequence length is set to 128, and the number of epochs is set to 3. The local PLM, the global PLM, and the pre-trained LLM are updated using the AdamW optimizer [67] with initial learning rates of $\eta_k = 2e-5$, $\eta = 2e-5$, and $\eta_{\mathcal{G}} = 3e-6$, respectively. Particularly, due to its huge parameter size, the query and value matrices of the pre-trained LLM on the cloud server is exclusively fine-tuned using low-rank adaptation (LoRA) [68] with a rank of 8, a scaling factor of 32, and a dropout rate of 0.1, while keeping the remaining parameters frozen to improve training efficiency and reduce computational overhead. To ensure fair comparisons, the size of the synthetic dataset N_{syn} is set to match the size of the public proxy dataset N_{pub} used by existing KD-based FL baselines, *i.e.*, 25,000 public samples from IMDB are used for baselines while 25,000 synthetic samples are generated for FedSTG in SST-2, 49,401 public samples from PAWS are used for baselines while 49,401 synthetic samples are generated for FedSTG in QQP, and 550,152 public samples from SNLI are used for baselines while 550,152 synthetic samples are generated for FedSTG in MNLI. Furthermore, to ensure reliable performance evaluation, the experiments on each benchmark task are run three times using different random seeds, and the average performance together with standard deviation of the global PLM on the development set of each dataset are reported to measure the effectiveness of different KD-based FL frameworks.

D. Main Results

Evaluation on Homogeneous Scenarios: Table II presents the performance comparison between FedSTG and existing KD-based FL frameworks on benchmark tasks under homogeneous FL scenarios. Overall, FedSTG consistently

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TABLE I
PROMPT TEMPLATES AND GENERATION CONFIGURATIONS ON BENCHMARK TASKS

Hyperparameter	Sentiment Analysis	Paraphrase Detection	Natural Language Inference
verbalizer	0: negative 1: positive	0: unequal 1: equivalent	0: entailed 1: neutral 2: contradictory
template	You are a professional synthetic text generator for sentiment analysis tasks. Please generate a [LABEL_WORD] movie review without any extra explanation.	You are a professional synthetic text generator for paraphrase detection tasks. Please generate two semantically [LABEL_WORD] sentences separated by "|" without any extra explanation.	You are a professional synthetic text generator for natural language inference tasks. Please generate a premise sentence and a [LABEL_WORD] hypothesis sentence separated by "|" without any extra explanation.
max_new_tokens	100	100	100
do_sample	True	True	True
top_k	50	50	50
top_p	0.9	0.9	0.9
temperature	0.8	0.8	0.8
no_repeat_ngram_size	3	3	3
num_return_sequences	1	1	1

TABLE II
PERFORMANCE EVALUATION ON BENCHMARK TASKS UNDER HOMOGENEOUS SCENARIOS

Method	Sentiment Analysis		Paraphrase Detection		Natural Language Inference	
	SST2→IMDB		QQP→PAWS		MNLI→SNLI	
	Acc		F1/Acc		Acc m/mm	
	α=100	α=1	α=100	α=1	α=100	α=1
model parameter sharing						
FedED [19]	89.0±1.1	86.8±1.2	84.9±0.8/87.5±0.6	83.3±0.8/85.7±1.2	80.2±0.5/80.8±0.6	78.0±1.0/78.7±0.8
FedDF [20]	**90.8±0.7**	89.2±0.9	**87.2±0.4/89.6±0.5**	84.5±0.6/86.4±0.7	**81.6±0.5/81.9±0.5**	80.2±0.8/80.5±0.7
model prediction sharing						
MHAT [64]	87.2±1.0	85.8±1.3	83.3±0.6/85.7±0.8	81.5±1.0/84.0±0.9	78.4±0.8/78.6±1.2	77.6±1.4/77.7±1.3
DS-FL [65]	86.8±1.2	85.3±1.4	82.7±0.5/85.3±1.2	80.9±0.9/83.6±1.0	78.0±1.1/78.1±1.2	77.0±1.3/77.2±0.9
FedID [21]	87.9±0.8	86.4±1.1	84.1±0.5/86.8±0.5	82.7±0.3/85.3±0.8	79.6±0.4/80.3±0.5	78.5±0.7/79.2±1.2
FedSTG	90.0±0.6	**88.4±0.9**	86.7±0.3/88.9±0.4	**84.8±0.8/86.5±0.9**	81.2±0.6/81.3±0.7	**80.6±1.1/80.8±1.1**

outperforms the baselines built upon model prediction sharing by a clear performance gap, and even approaches the baselines built upon model parameter sharing, which can be attributed to its unique ability to generate domain-specific synthetic data for effective knowledge transfer. In contrast, existing KD-based FL baselines suffer from a considerable domain gap between public proxy data and private local data. This mismatch severely limits their capacity to transfer domain-specific knowledge, resulting in suboptimal performance. Furthermore, although FedED and MHAT explicitly incorporate label information from the public proxy dataset to provide additional supervision while FedID utilizes external annotated data to mitigate confirmation bias, none of them can deliver noticeable performance gains due to the underlying domain gap remains persistent. Instead, FedSTG leverages the domain-specific knowledge encoded in local PLMs distributed across user devices to guide the pre-trained LLM on the cloud server in generating high-quality synthetic text, which closely aligns with the underlying private domain and thus substantially narrows the domain gap, leading to more effective knowledge transfer. Notably, the performance differences between FedSTG and existing KD-based FL baselines become more pronounced when encountering non-IID data distributions across user devices, which further highlights its superior ability to bridge the domain gap.

Evaluation on Heterogeneous Scenarios: Consistent with the design philosophy of MHAT, DS-FL, and FedID, FedSTG replaces the aggregation of model parameters with KD, enabling knowledge transfer between the cloud server and user devices without sharing any model parameters. This paradigm enhances its versatility in real-world applications by naturally supporting both homogeneous and heterogeneous FL scenarios. Table III presents the performance comparison between FedSTG and existing KD-based FL frameworks on benchmark tasks under heterogeneous FL scenarios. It can be observed that FedSTG consistently achieves superior performance compared to all baselines, exhibiting the same trend as that observed in homogeneous FL scenarios. These findings underscore the adaptability and robustness of FedSTG across diverse FL scenarios, demonstrating its great potential in serving as a unified solution for both homogeneous and heterogeneous FL.

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TABLE III
PERFORMANCE EVALUATION ON BENCHMARK TASKS UNDER HETEROGENEOUS SCENARIOS

Method	Sentiment Analysis		Paraphrase Detection		Natural Language Inference	
	SST2→IMDB		QQP→PAWS		MNLI→SNLI	
	Acc		F1/Acc		Acc m/mm	
	α=100	α=1	α=100	α=1	α=100	α=1
MHAT [64]	86.7±0.9	85.3±1.0	82.8±0.6/85.2±0.8	81.2±0.7/83.6±1.2	78.0±0.7/78.1±0.8	77.2±0.5/77.3±0.9
DS-FL [65]	86.0±0.7	84.9±0.9	82.5±0.6/84.7±1.0	80.8±0.8/83.0±1.0	77.6±1.1/77.7±1.1	76.5±0.3/76.7±0.4
FedID [21]	87.6±1.1	86.0±1.3	83.6±0.5/86.3±0.7	82.3±0.3/84.9±0.4	78.9±0.9/78.8±1.3	78.1±1.0/78.8±1.1
FedSTG	**89.3±0.6**	**88.0±0.8**	**86.0±0.3/88.5±1.0**	**84.4±0.4/86.2±0.6**	**80.8±0.7/81.0±0.8**	**80.2±0.5/80.5±0.8**

TABLE IV
ABLATION STUDY ON BENCHMARK TASKS

Method	Sentiment Analysis		Paraphrase Detection		Natural Language Inference	
	SST2→IMDB		QQP→PAWS		MNLI→SNLI	
	Acc		F1/Acc		Acc m/mm	
	α=100	α=1	α=100	α=1	α=100	α=1
FedSTG	**90.0±0.6**	**88.4±0.9**	**86.7±0.3/88.9±0.4**	**84.8±0.8/86.5±0.9**	**81.2±0.6/81.3±0.7**	**80.6±1.1/80.8±1.1**
w/o Weight-based Ensemble	89.5±1.0	87.7±1.3	86.2±0.5/88.3±0.8	83.8±0.4/85.7±0.7	80.5±0.5/80.7±0.5	79.8±0.7/80.1±1.3
w/o Multi-round Refinement	86.8±0.9	85.1±1.3	82.9±0.9/85.0±1.0	81.2±0.8/83.8±1.1	78.3±0.5/78.5±1.0	77.6±0.8/77.8±0.9
+Zero-shot Prompting	82.6±1.1	80.8±1.3	78.3±1.0/80.5±1.2	76.1±0.4/78.4±0.7	73.1±1.2/73.4±1.2	71.9±1.0/72.4±1.3
+Few-shot Prompting	84.8±0.6	83.0±1.0	80.2±0.3/82.3±1.0	78.7±0.9/80.3±1.2	75.4±0.6/76.0±1.0	74.2±0.3/74.6±0.5
+Well-designed Prompting	85.7±0.9	83.8±1.1	81.4±0.5/83.1±0.6	78.9±0.3/81.8±0.7	76.7±0.7/77.1±1.1	75.4±0.6/75.5±1.0

E. Ablation Study

Effect of Weight-based Ensemble: To evaluate the contribution of the weight-based ensemble mechanism in FedSTG, we conduct an ablation study by replacing it with a simple average-based ensemble mechanism, where the model predictions from locally trained PLMs are directly averaged using identical weights. As shown in the results of "w/o Weight-based Ensemble" in Table IV, this modification leads to a certain degree of performance drops across benchmark tasks. Intuitively, user devices that host larger volumes of private data tend to produce more reliable model predictions due to sufficient local training, and assigning them proportionally higher weights yields a more faithful estimation of the underlying data distribution. In contrast, uniform averaging treats all devices equally, making the aggregated model predictions more susceptible to noise from under-trained or biased devices with limited private data.

Effect of Multi-round Refinement: To evaluate the contribution of the multi-round refinement mechanism in FedSTG, we conduct an ablation study by replacing it with a single-round refinement mechanism, in which the pre-trained LLM is updated only once using the rewards provided by local PLMs distributed across user devices. As shown in the results of "w/o Multi-round Refinement" in Table IV, performing only a single round of refinement results in noticeable performance sacrifices across benchmark tasks as it fails to provide the LLM with enough domain-specific knowledge for adaptation. In contrast, iteratively aligning the pre-trained LLM with locally trained PLMs enables progressive improvements in generation quality to narrow the domain gap between the generated synthetic dataset and the underlying private dataset, stabilizing the subsequent knowledge transfer process.

TABLE V
GENERATION EVALUATION ON BENCHMARK TASKS

Task		Similarity↑	PII↓
Sentiment Analysis	SST-2→IMDB	0.5645	38.45%
	SST-2→Synthetic	**0.6489**	**15.21%**
Paraphrase Detection	QQP→PAWS	0.7254	28.76%
	QQP→Synthetic	**0.8223**	**10.73%**
Natural Language Inference	MNLI→SNLI	0.6554	21.45%
	MNLI→Synthetic	**0.7831**	**6.84%**

Effect of Collaborative Prompting: To evaluate the advantages of the proposed collaborative prompting strategy in FedSTG, where the generation capability of the pre-trained LLM is collaboratively enhanced by the domain-specific understanding capabilities of local PLMs distributed across user devices, we conduct an ablation study by replacing it with three state-of-the-art prompting strategies: 1) Zero-shot prompting [5], which directly prompts the pre-trained LLM with task and label descriptions without any domain adaptation, 2) Few-shot prompting [5], which provides the pre-trained LLM with ten demonstrations sampled from the private dataset to steer its subsequent generation, and 3) Well-designed prompting [35], which augments the prompt template with additional details about domain characteristics. The empirical results presented in Table IV reveal that the collaborative prompting strategy substantially outperforms alternative prompting strategies due to domain-specific adaptation on the pre-trained LLM is conducive to eliciting its generation capability compared to prompting engineering. Particularly, although the few-shot prompting strategy directly exploits private local data while the well-designed prompting strategy explicitly

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TABLE VI
EFFICIENCY EVALUATION ON BENCHMARK TASKS

Method	Task	GPU	Runtime	Downlink	Uplink
MHAT [64]	Sentiment Analysis	1 h	1.2 h	$\left\|\mathcal{D}_{pub}\right\| \cdot K \cdot 1 + \left\|\mathcal{F}(\mathcal{D}_{pub}; \Theta^t)\right\| \cdot K \cdot T$	$\left\|\mathcal{F}_k(\mathcal{D}_{pub}; \Theta_k^t)\right\| \cdot K \cdot T$
	Paraphrase Detection	3 h	3.4 h		
	Natural Language Inference	4 h	4.5 h		
DS-FL [65]	Sentiment Analysis	1 h	1.1 h	$\left\|\mathcal{D}_{pub}\right\| \cdot K \cdot 1 + \left\|\sum_{k \in \mathcal{K}^t} \frac{N_k}{N} \mathcal{F}_k(\mathcal{D}_{pub}; \Theta_k^t)\right\| \cdot K \cdot T$	$\left\|\mathcal{F}_k(\mathcal{D}_{pub}; \Theta_k^t)\right\| \cdot K \cdot T$
	Paraphrase Detection	3 h	3.2 h		
	Natural Language Inference	4 h	4.3 h		
FedID [21]	Sentiment Analysis	1 h	1.3 h	$\left\|\mathcal{D}_{pub}\right\| \cdot K \cdot 1 + \left\|\sum_{k \in \mathcal{K}^t} \frac{N_k}{N} \mathcal{F}_k(\mathcal{D}_{pub}; \Theta_k^t)\right\| \cdot K \cdot T$	$\left\|\mathcal{F}_k(\mathcal{D}_{pub}; \Theta_k^t)\right\| \cdot K \cdot T$
	Paraphrase Detection	3 h	3.5 h		
	Natural Language Inference	4 h	4.7 h		
FedSTG	Sentiment Analysis	2 h	2.5 h	$\left\|\mathcal{D}_{syn}\right\| \cdot K \cdot T + \left\|\sum_{k \in \mathcal{K}^t} \frac{N_k}{N} \mathcal{F}_k(\mathcal{D}_{syn}; \Theta_k^t)\right\| \cdot K \cdot T$	$\left\|\mathcal{F}_k(\mathcal{D}_{syn}; \Theta_k^t)\right\| \cdot K \cdot T$
	Paraphrase Detection	5 h	5.7 h		
	Natural Language Inference	7 h	7.9 h		

incorporates domain characteristics, the risk of privacy violation is not traded for significant performance gains compared to the privacy-preserving collaborative prompting strategy.

F. Qualitative Analysis

Analysis of Generation Quality: To assess the generation quality of synthetic text, we measure the cosine similarity between the private dataset and the public proxy dataset, as well as the cosine similarity between the private dataset and the synthetic dataset generated by FedSTG. Specifically, these three datasets are first encoded into embedding vectors using all-MiniLM-L6-v2 [69] from the sentence-transformers library [70]. Subsequently, the cosine similarity between the averaged embedding representations of any two datasets is summarized in Table V. These numerical results show that the cosine similarity between the private dataset and the synthetic dataset is significantly higher than that between the private dataset and the public proxy dataset. This indicates that the synthetic dataset generated by FedSTG more closely aligns with the underlying distribution of the private domain, effectively narrowing the domain gap that limits the performance of existing KD-based FL frameworks.

Analysis of Data Privacy: To assess the potential risk associated with disclosing private information such as personally identifiable information (PII) in the synthetic dataset generated by FedSTG compared to the public proxy dataset, we employ Presidio,[3] a data protection and de-identification tool provided by Microsoft, to detect each sample whether it contains privacy-sensitive entities, including names, locations, phone numbers, financial data, medical information, and other identifiable attributes. Subsequently, the proportion of samples containing such privacy-sensitive entities relative to the total number of samples in each dataset is summarized in Table V. These numerical results show that the synthetic dataset generated by FedSTG contains significantly fewer privacy-sensitive entities than the public proxy dataset, which can be attibuted to the former is generated from pseudo data while the latter is curated from user-generated real data, which often inadvertently includes sensitive information. This indicates that FedSTG not only produces domain-aligned synthetic text but also avoids replicating identifiable patterns from the private domain, thereby reducing the likelihood of exposing private information.

Analysis of Computational Cost: To quantify the computational costs across different KD-based FL frameworks, we record the average GPU hours and runtime per round in Table VI. The results indicate that FedSTG incurs higher computation and latency compared to other KD-based FL baselines due to the integration of a pre-trained LLM, which exclusively introduces a data generation stage and a reward-guided adaptation stage. Nevertheless, this computational cost remains acceptable for two reasons. First, both stages are executed only once per round on the cloud server, without increasing the computational burdens on resource-constrained user devices. Second, the incorporation of LoRA significantly reduces the number of trainable parameters, achieving parameter-efficient adaptation while maintaining model performance.

Analysis of Communication Overhead: To provide a comprehensive comparison of communication overhead across different KD-based FL frameworks, we summarize the formulations of both downlink and uplink communication overhead in Table VI. As indicated, FedSTG introduces additional downlink communication overhead compared to other KD-based FL baselines due to the synthetic dataset generated in the previous round needs to be replaced by the newly generated one for multi-round refinement. This results in extra $T-1$ rounds of downlink transmissions for each device to download the updated synthetic dataset from the cloud server. Nonetheless, this communication overhead is acceptable due to the bandwidth provided by the downlink communication is usually much higher than that of the uplink communication in real-world environments, making the incremental overhead relatively minor.

G. Quantitative Analysis

Impact of Model Size: To evaluate the robustness of FedSTG under different scales of LLMs, we investigate the performance of FedSTG across benchmark tasks with Llama-3.2-1B-Instruct, Llama-3.2-3B-Instruct, and Llama-3.1-8B-Instruct while keeping all other experimental settings consistent. As shown in Fig. 3, the performance of FedSTG improves with the introduction of high-capacity LLMs due to the fact that larger-scale models

[3] https://github.com/microsoft/presidio/

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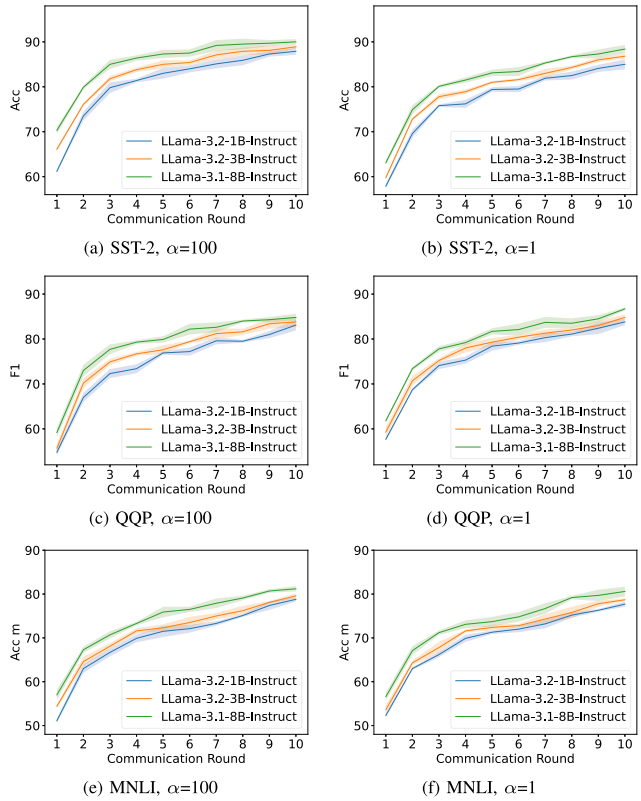
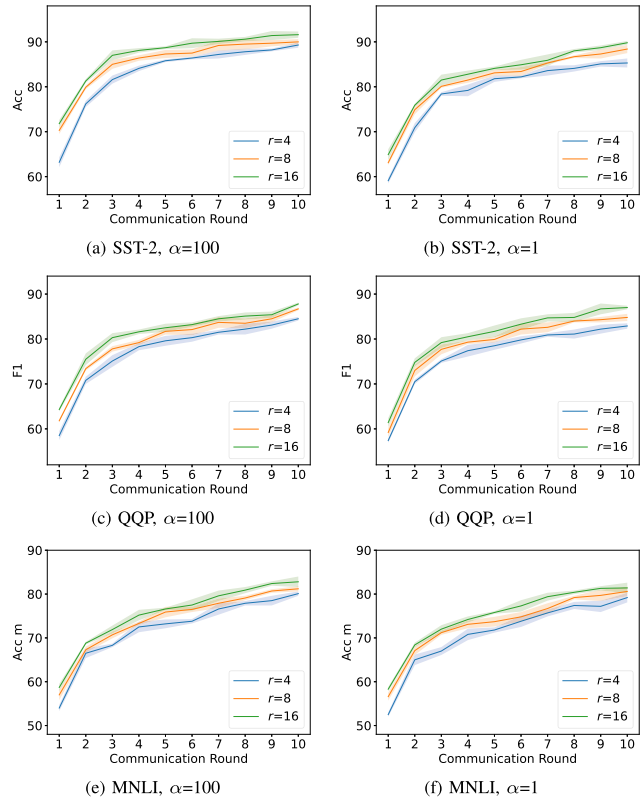


(a) SST-2, α=100 (b) SST-2, α=1
(c) QQP, α=100 (d) QQP, α=1
(e) MNLI, α=100 (f) MNLI, α=1

Fig. 3. Performance evaluation on benchmark tasks with varying model sizes.



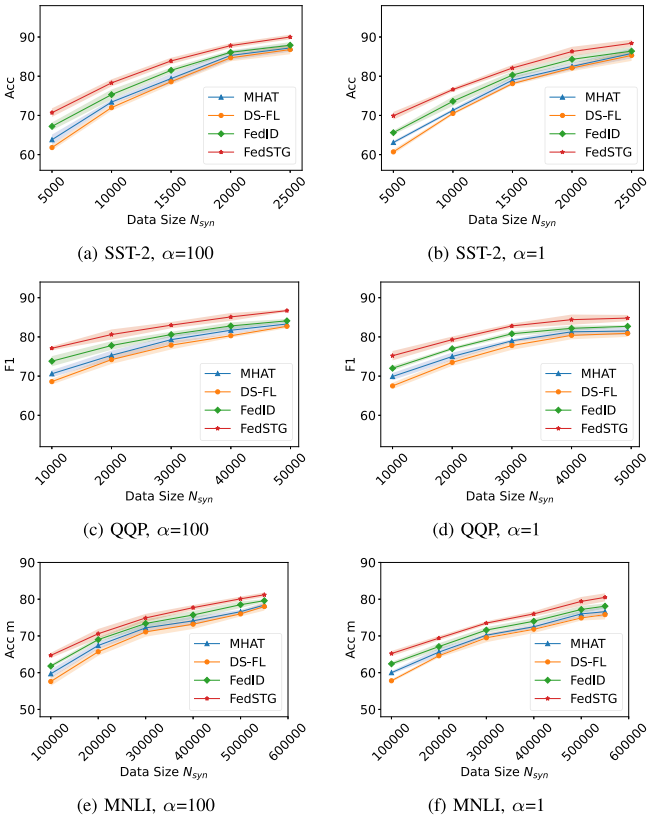
(a) SST-2, α=100 (b) SST-2, α=1
(c) QQP, α=100 (d) QQP, α=1
(e) MNLI, α=100 (f) MNLI, α=1

Fig. 4. Performance evaluation on benchmark tasks with varying LoRA ranks.

typically are endowed with stronger generation capabilities to further narrow the domain gap between the synthetic dataset and the private dataset.

Impact of LoRA Rank: To evaluate the robustness of FedSTG under different configurations of LoRA ranks, we investigate the performance of FedSTG across benchmark tasks with varying LoRA ranks $r \in \{4, 8, 16\}$ while keeping all other experimental settings consistent. As shown in Fig. 4, increasing the LoRA rank consistently leads to performance improvements due to higher-rank LoRA updates a larger parameter space to enhance the adaptation capacity of the LLM during reward-guided fine-tuning. However, higher LoRA ranks also introduce higher computational costs and longer training time. To strike a balance between model performance and training efficiency, we adopt a default LoRA rank of 8.

Impact of Data Size: To evaluate the robustness of FedSTG under different sizes of synthetic datasets, we investigate the performance of FedSTG across benchmark tasks with varying data sizes. Specifically, for SST-2, we vary the data size N_{syn} from 5,000 to 25,000 in increments of 5,000. For QQP, we vary the data size N_{syn} from 10,000 to 49,401 in increments of 10,000. For MNLI, we vary the data size N_{syn} from 100,000 to 550,152 in increments of 100,000. Note that the size of the synthetic dataset N_{syn} in FedSTG is equal to the size of the public proxy dataset N_{pub} in existing KD-based FL baselines for fair comparisons. As shown in Fig. 5, all frameworks initially exhibit rapid performance improvements as the data size increases, eventually reaching a plateau. However, FedSTG consistently outperforms existing KD-based FL baselines across all data



(a) SST-2, α=100 (b) SST-2, α=1
(c) QQP, α=100 (d) QQP, α=1
(e) MNLI, α=100 (f) MNLI, α=1

Fig. 5. Performance evaluation on benchmark tasks with varying data sizes.

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TABLE VII
EXAMPLE ANALYSIS ON BENCHMARK TASKS

Task	Label	Private Local Data	Public Proxy Data	Synthetic Data
Sentiment Analysis	negative	At three hours and with very little story or character development, there is plenty of room for editing, and a much shorter cut surely would have resulted in a smoother, more focused narrative without sacrificing any of the cultural intrigue.	Ned aKelly is such an important story to Australians but this movie is awful. It's an Australian story yet it seems like it was set in America. Also Ned was an Australian yet he has an Irish accent...it is the worst film I have seen in a long time.	I walked out of the theater feeling completely underwhelmed by the latest blockbuster. The plot was a jumbled mess of cliches and overused tropes, with characters that felt more like cardboard cutouts than actual people. The dialogue was cringe-worthy and often sounded like it was written by a middle schooler trying to sound cool. The "twist" at the end was predictable and didn't even come close to surprising me.
	positive	Swimming is above all about a young woman's face, and by casting an actress whose face projects that woman's doubts and yearnings, it succeeds.	This is the definite Lars von Trier Movie, my favorite, I rank it higher than "Breaking the waves" or the latest "Dancer in the Dark"...I simply love the beauty of the picture...the framing is so original; acting is wonderful, AMUST SEE.	An absolute must-see! The chemistry between the leads is undeniable, and the direction is masterful. The pacing is expertly handled, with just the right amount of tension and release. The special effects are seamless, and that iconic scene will leave you breathless. A true classic that will be cherished for generations to come.
Paraphrase Detection	not_equivalent	*A*: How is the life of a math student? Could you describe your own experiences? *B*: Which level of preparation is enough for the exam jlpt5?	*A*: The Soviet Union maintained an embassy in Oslo and a consulate in Barentsburg, while Norway maintained a message in Moscow. *B*: The Soviet Union maintained an embassy in Moscow and a consulate in Barentsburg, while Norway maintained a message in Oslo.	*A*: The artist's use of vibrant colors in her painting reflected her emotional state, which was characterized by a sense of calmness and serenity. *B*: The artist's painting was a chaotic mess of colors that seemed to scream with anxiety and frustration.
	equivalent	*A*: Why did it take so long for NASA to find water on Mars? *B*: How much chances are there that NASA already knew that there is water on Mars?	*A*: Briggs later met Briggs at the 1967 Monterey Pop Festival, where Ravi Shankar was also performing, with Eric Burdon and The Animals. *B*: Briggs met Briggs later at the Monterey Pop Festival of 1967, where Ravi Shankar also performed with Eric Burdon and The Animals.	*A*: The increasing trend of using renewable energy sources is expected to significantly reduce green house gas emissions. *B*: The usage of renewable energy resources is projected to lead to a substantial decrease in the release of greenhouse gases into the atmosphere.
Natural Language Inference	entailment	*A*: Clark also expressed the hope that he and Redgrave could continue with their marriage. *B*: Clark hoped that he could continue their marriage.	*A*: A man and a woman cross the street in front of a pizza and gyro restaurant. *B*: Near a couple of restaurants, two people walk across the street.	*A*: The new employee struggled to adapt to the company's complex software systems, resulting in a significant decrease in productivity. *B*: Therefore, the company should consider providing additional training or support to help the employee better understand the software.
	neutral	*A*: In the midst of this amazing amalgam of cultures is a passion for continuity. *B*: A passion for continuity is not the most important of these cultures.	*A*: People waiting to get on a train or just getting off. *B*: The people waiting on the train are sitting.	*A*: The rise in popularity of veganism has led to an increased demand for plant-based alternatives in the food industry. *B*: There is a correlation between the growth of the veganism movement and the increase in plant-based food options available in the market.
	contradiction	*A*: This analysis pooled estimates from these two studies to develop a C-R function linking PM to chronic bronchitis. *B*: The analysis proves that there is no link between PM and bronchitis.	*A*: A woman is walking across the street eating a banana, while a man is following with his briefcase. *B*: A woman eats ice cream walking down the sidewalk, and there is an other woman in front of her with a purse.	*A*: The new AI system has been trained on a vast amount of data and is capable of learning at an exponential rate. *B*: The new AI is actually a sophisticated program that can only perform basic calculations.

sizes, and its performance advantage becomes more pronounced under data-scarce conditions. This observation suggests that the synthetic dataset generated by FedSTG effectively bridges the domain gap between the public proxy dataset and the private dataset, facilitating effective knowledge transfer.

Impact of Sampling Fraction: To evaluate the robustness of FedSTG under different scales of participants, we investigate the performance of FedSTG across benchmark tasks with varying sampling fractions $q \in \{0.1, 0.5, 1\}$. As shown in Fig. 6, the performance of all frameworks improves as the sampling fraction increases due to the fact that the involvement of more participating devices results in more private data being able to be exploited for performance improvements. However, FedSTG consistently maintains superior performance compared to existing KD-based FL baselines across a range of sampling fractions, demonstrating robust scalability and stability in real-world FL scenarios.

H. Case Study

Example Analysis: We present several representative examples sampled from the private dataset, the public proxy dataset,

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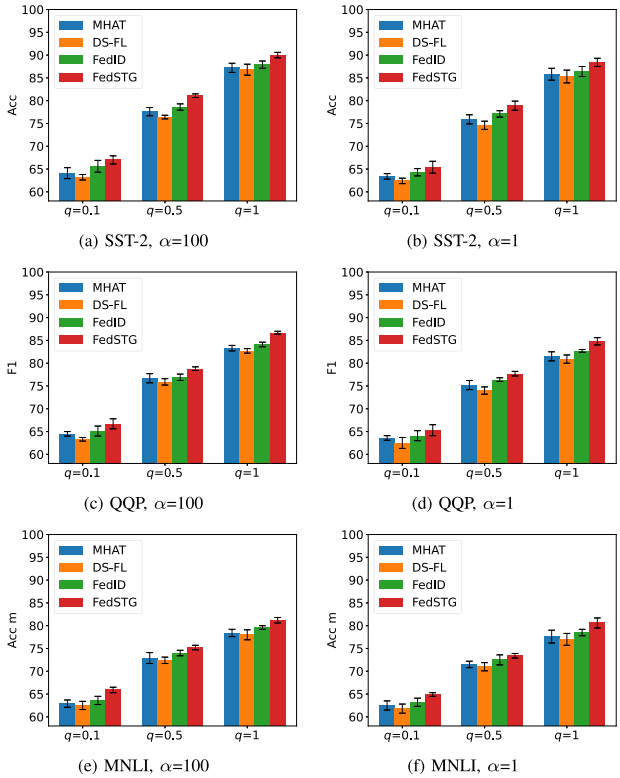
TABLE VIII
ERROR ANALYSIS ON BENCHMARK TASKS

Task	Input	Label	Prediction	Failure Mode
Sentiment Analysis	The emotions are raw and will strike a nerve with anyone who's ever had family trauma.	positive	negative	Label Ambiguity
	Pumpkin takes an admirable look at the hypocrisy of political correctness , but it does so with such an uneven tone that you never know when humor ends and tragedy begins.	negative	positive	Domain Shift
Paraphrase Detection	*A*: How is India preparing for war against Pakistan? *B*: Is India preparing for a war?	not_equivalent	equivalent	Reward Misalignment
	A: "What does a good answer on Quora look like? What does it mean to ""be helpful""?" *B*: How do you write a good answer on Quora?	equivalent	not_equivalent	Label Ambiguity
Natural Language Inference	*A*: Of those beginning prenatal care in the first trimester, Indiana mothers rank in the lower third of individuals receiving such care nationwide. *B*: There are Indiana mothers who do not receive prenatal care in the first trimester.	entailment	neutral	Label Ambiguity
	A: Are you and since being Argentinean we also have a lot of pasta *B*: Argentineans are fond of pasta because many of its citizens are Italian descendants.	neutral	contradiction	Reward Misalignment
	A: Not quite as large is the Papal Crose commemorating Pope John Paul II's visit in 1979, when more than one million people gathered to celebrate mass. *B*: Pope John Paul II also visited in 1983.	neutral	entailment	Reward Misalignment
	A: For big Raj-buffs, the supreme example of Indo-Gothic style is the Victoria Terminus, affectionately abbreviated to VT nowadays, once the railway station that launched adventures inland, now handling mostly suburban traffic. *B*: The Victoria Terminus is still the main departure point for travelers heading deeper into the country.	contradiction	neutral	Domain Shift



(a) SST-2, α=100 (b) SST-2, α=1
(c) QQP, α=100 (d) QQP, α=1
(e) MNLI, α=100 (f) MNLI, α=1

Fig. 6. Performance evaluation on benchmark tasks with varying sampling fractions.

and the synthetic dataset for each benchmark task in Table VII. As indicated, the synthetic data generated by FedSTG exhibits strong semantic coherence and contextual relevance, closely aligning with the linguistic structures and semantic patterns in private local data. These examples further highlight the ability of FedSTG to generate high-quality synthetic data, which serve as a reliable medium for knowledge transfer between the cloud server and user devices, eliminating the need to curate a public proxy dataset that requires prior knowledge about the private domain.

Error Analysis: We present several representative misclassification cases produced by FedSTG for each benchmark task in Table VIII, which reveal the presence of several failure modes in FedSTG. First, label ambiguity occurs when subtle semantic cues are insufficiently captured in the generated synthetic dataset, and the subsequent knowledge transfer causes the global PLM to misinterpret nuanced instances. Second, domain shift persists due to certain domain-specific linguistic patterns are underrepresented in the generated synthetic dataset, thereby limiting the capability of the global PLM to generalize to real-world user-generated data. Third, reward misalignment arises from misleading reward signals from local PLMs across user devices, which potentially induces the LLM to produce overconfident yet low-quality synthetic data for knowledge transfer. These observations suggest that improving the fidelity and diversity of synthetic text is essential for further advancing the generalization of FedSTG.

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V. Conclusions and Limitations

This paper presents FedSTG, a privacy-preserving FL framework for NLP empowered by LLM-driven synthetic text generation. By combining the exceptional generation capability of a pre-trained LLM deployed on the cloud server with the domain-specific understanding capabilities of local PLMs distributed across user devices, FedSTG generates high-quality synthetic text for knowledge transfer, facilitating the integration of KD and FL without the need for publicly available proxy data. Empirical results with extensive analysis across a wide range of benchmark tasks demonstrate the superiority of FedSTG compared to existing KD-based FL frameworks.

However, FedSTG also presents several new challenges. First, it relies on high-capacity LLMs for faithful interpretation of task instructions and friendly controllability over output formats, which is beneficial for task-oriented synthetic text generation. Second, its scalability in generating large-scale synthetic corpora may be constrained by the inefficient generation and limited diversity inherent in autoregressive sampling. Third, it remains susceptible to potential privacy risks under an honest-but-curious threat model, which may launch malicious attacks such as membership inference attacks or data reconstruction attacks to analyze the synthetic dataset. As a promising direction, future work will explore differentially private synthetic text generation for enhanced privacy guarantees.

Acknowledgment

The authors would like to thank the anonymous reviewers for their constructive comments.

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